

# Glory Aditya Jauzaulbahi

✉ gloryadityaj@gmail.com | ☎ +62-858-9537-7701  
🌐 linkedin.com/in/gloryadityaj | 🏠 github.com/adityaglor

## PROFESSIONAL SUMMARY

AI and ML Data Engineer with a proven track record in designing robust predictive models, training deep learning architectures, and architecting scalable distributed data systems. Adept at transforming complex multimodal datasets spanning 1D time series signals to 2D high resolution imagery data into high performing and low latency solutions. Brings applied expertise in advanced generative AI, MLOps, and model deployments to bridge the gap between mathematical modeling and production grade software engineering.

## TECHNICAL SKILLS

**Software Dev & Tools:** Python, SQL (PostgreSQL), Git/GitHub, Linux (Ubuntu/Raspbian).  
**Data Engineering & Architecture:** Apache Kafka, PySpark, Apache Airflow, ChromaDB.  
**AI Modeling & Computer Vision:** PyTorch, TensorFlow, Scikit learn, LangChain, ONNX, OpenCV, MediaPipe, EasyOCR, Roboflow.  
**MLOps & CI/CD:** Docker, FastAPI, MLFlow, EvidentlyAI, Streamlit, GitHub Actions, Prometheus, Grafana, Pytest, Microsoft Azure AI.

## EDUCATION

**Universitas Gadjah Mada** Yogyakarta, Indonesia  
*Bachelor of Engineering in Biomedical Engineering* Graduated: Aug 2025  
• **Focus Areas:** Signal & Image Processing, Bioinformatics, Artificial Intelligence, Intelligence System.

**Pacmann AI** Online  
*AI/ML Engineering Bootcamp* Jan 2024 - March 2026  
• **Focus Areas:** Data Visualization, Probability & Statistics, Machine Learning Process, Recommender Sytem, Model Deployment & MLOps, NLP & LLM.

## PROFESSIONAL EXPERIENCE

**Pusat Riset Sains Data & Informasi, BRIN** Bandung, Indonesia  
*AI Researcher Intern* Jul 2022 - Sep 2022

- Curated and acquired dataset from manual recording and diverse online resources, programmatically rendering over 7,000 frames dataset for robust gesture recognition.
- Engineered a robust computer vision pipeline utilizing MediaPipe to extract temporal features from video datasets, establishing an automated data flow for gesture recognition.
- Deployed and optimized SVM models for edge computing, achieving real-time hand motion classification with 1 to 4 FPS model inference on Raspberry Pi 4 across varied environmental conditions.

🏠 Project Repository

## PROJECTS EXPERIENCE

**Meltion (Melanoma Detection) Android App – DTETI Capstone Team Project**

- Architected a scalable preprocessing workflow for over 10,000 diverse skin lesion images by implementing robust illumination correction and data augmentation strategies.
- Fine-tuned a ResNet50 model to achieve 78% accuracy in multiclass detection while maximizing recall for critical diagnostic reliability.

- Collaborated on backend integration and model deployment via Google Vertex AI to enable seamless low-latency inference for mobile end users.

### **Data Augmentation for Imbalanced Dataset with GANs – Undergraduate Thesis**

- Developed custom GAN and VAE architectures to directly solve severe class imbalance and data scarcity in medical datasets.
- Synthesized high-resolution medical data and benchmarked perceptual realism to achieve a high quality medical data with Fréchet Inception Distance (FID) score of 0.2249.
- Improved the diagnostic utility of synthetic data by retraining downstream classifiers, driving a massive improvement in F1 score to 96% over baseline models.

 Project Repository

### **Biomedical Research Local RAG System**

- Architected a context-aware RAG system using Llama 3.2 to extract actionable insights from unstructured biomedical literature while optimizing MMR and context window parameters.
- Containerized a scalable document retrieval microservice using Docker and ChromaDB to drastically reduce information retrieval latency.
- Implemented advanced vector search methodologies to accelerate data retrieval to under 2 seconds while maintaining strict semantic accuracy.

 Project Repository

### **End-to-End Nutritional OCR & Data Extraction Pipeline**

- Developed an automated OCR data extraction pipeline using fine-tuned YOLO and EasyOCR to eliminate manual data entry bottlenecks for noisy packaging imagery.
- Engineered a fuzzy Regex processing engine that successfully recovered 100% of missing packaging metadata via automated imputation.
- Deployed a highly scalable dual-interface system capable of seamless batch processing with strict validation logic to eliminate data hallucinations.

 Project Repository

### **Distributed Medical Image Streaming**

- Architected an event-driven microservices architecture utilizing Apache Kafka and Docker Compose to stream and process medical imagery in real-time.
- Engineered a decoupled PyTorch consumer leveraging an optimized ResNet18 model to enable asynchronous low-latency pneumonia detection.
- Deployed an interactive Streamlit dashboard for live clinical monitoring and integrated automated Pytest validation suites to guarantee production-grade reliability.

 Project Repository

## **CERTIFICATIONS**

### **Azure AI Fundamentals (AI 900)**

*Microsoft, Sep 2025*

- Validated understanding of computer vision workloads and cloud-based AI deployment strategies.

### **DeepLearning.AI GANs Specialization**

*Coursera, Aug 2023*

- Advanced specialization in generative modeling and image reconstruction techniques.

## **LEADERSHIP EXPERIENCE**

### **Unit Kegiatan Mahasiswa Perisai Diri**

*Universitas Gadjah Mada, 2020-2021*

- **Head of Internal Division (2021):** Directed organizational workflows and optimized internal resource allocation.
- **Inventory Division Supervisor (2020):** Managed asset tracking systems and resource accountability for all organizational activities.